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(54) **DISTRIBUTED LAND MOBILE RADIO ARCHITECTURES**

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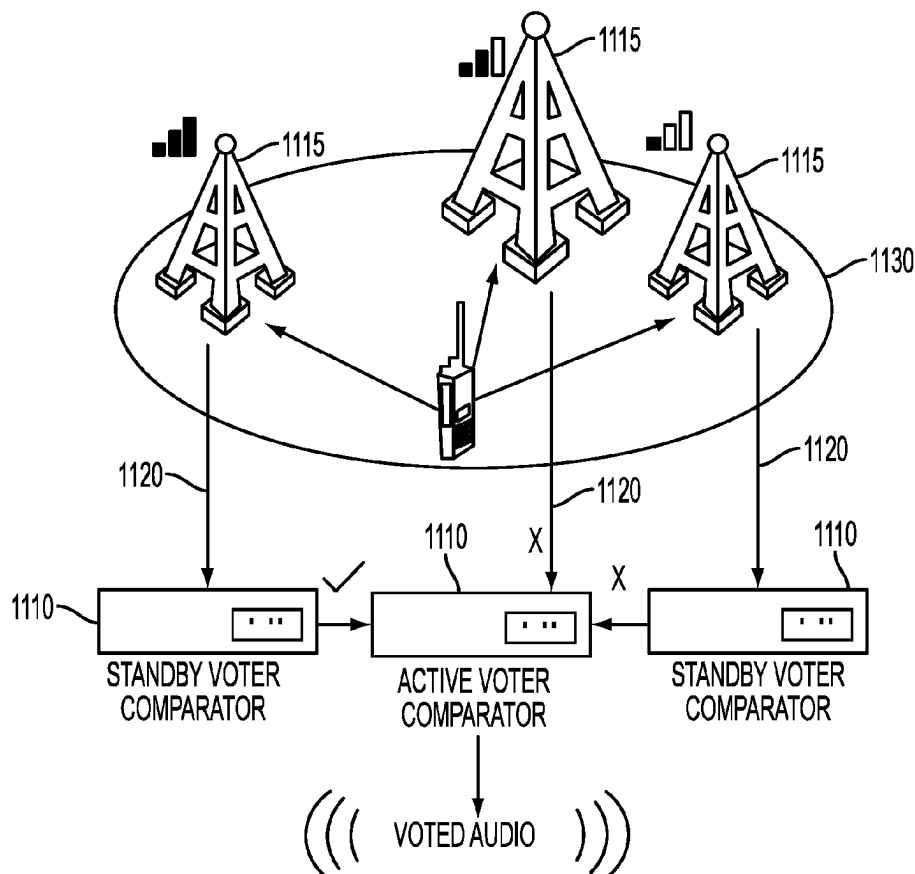
Related U.S. Application Data

(63) Continuation of application No. 18/595,143, filed on Mar. 4, 2024, now Pat. No. 12,316,437, which is a continuation of application No. 17/981,070, filed on Nov. 4, 2022, now Pat. No. 11,936,466, which is a continuation of application No. 17/100,213, filed on Nov. 20, 2020, now Pat. No. 11,496,212, which is a continuation of application No. 16/543,302, filed on Aug. 16, 2019, now Pat. No. 10,880,000, which is a continuation of application No. 15/494,391, filed on Apr. 21, 2017, now Pat. No. 10,461,846, which is a continuation of application No. 14/217,150, filed on Mar. 17, 2014, now Pat. No. 9,774,386.

(57)

ABSTRACT

A system for providing communication in a distributed land mobile radio (LMR) system architecture. In some embodiments, the system includes a first controller associated with an LMR site. The first controller may be configured to control communication, via a communication channel, of a plurality of LMRs. In some embodiments, the system further includes a first repeater of a plurality of repeaters associated with the LMR site. The first repeater may include at least an active mode. The first repeater may be in the active mode to initiate a simulcast controller operation using the communication channel of the plurality of repeaters.



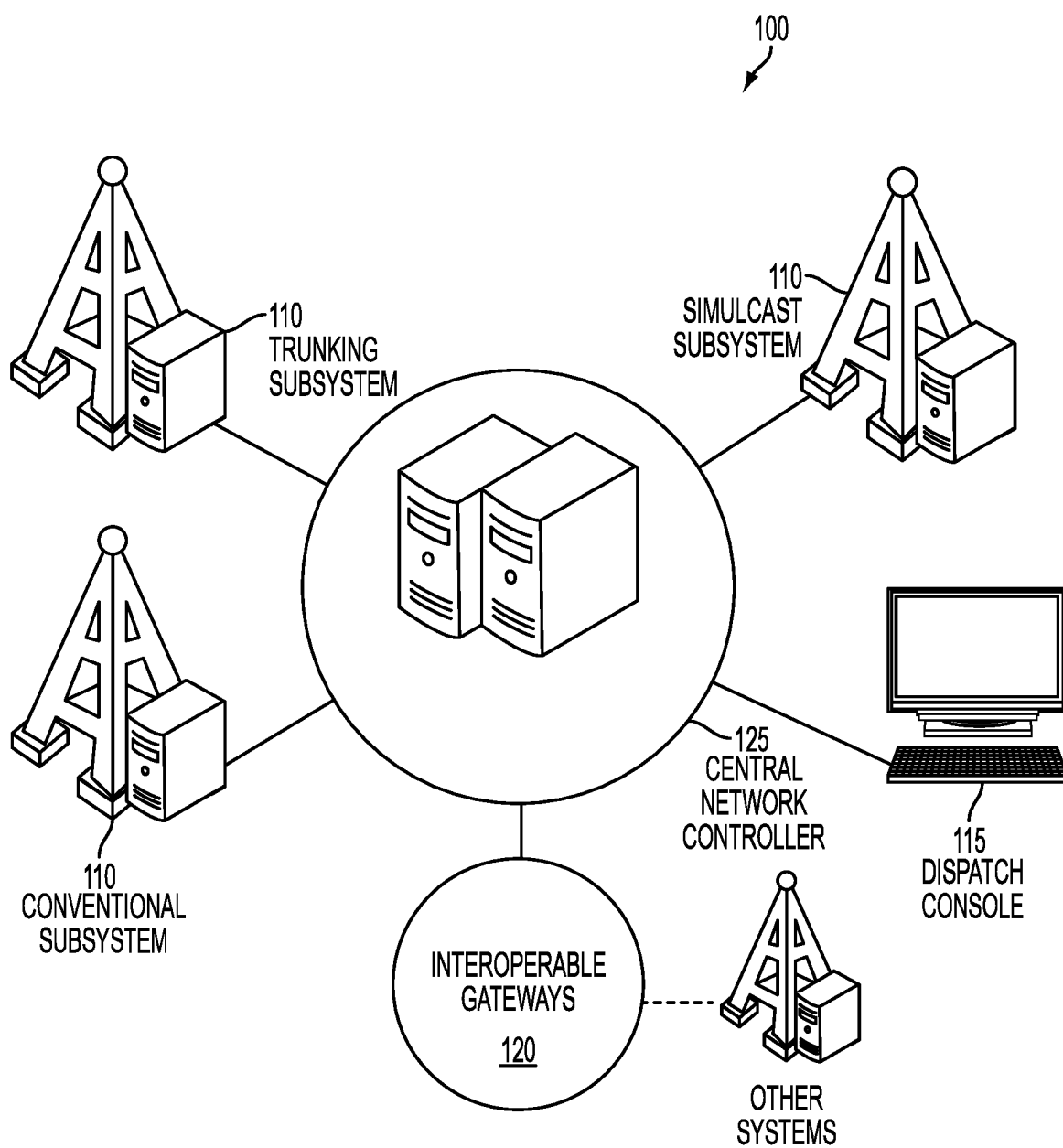


FIG. 1

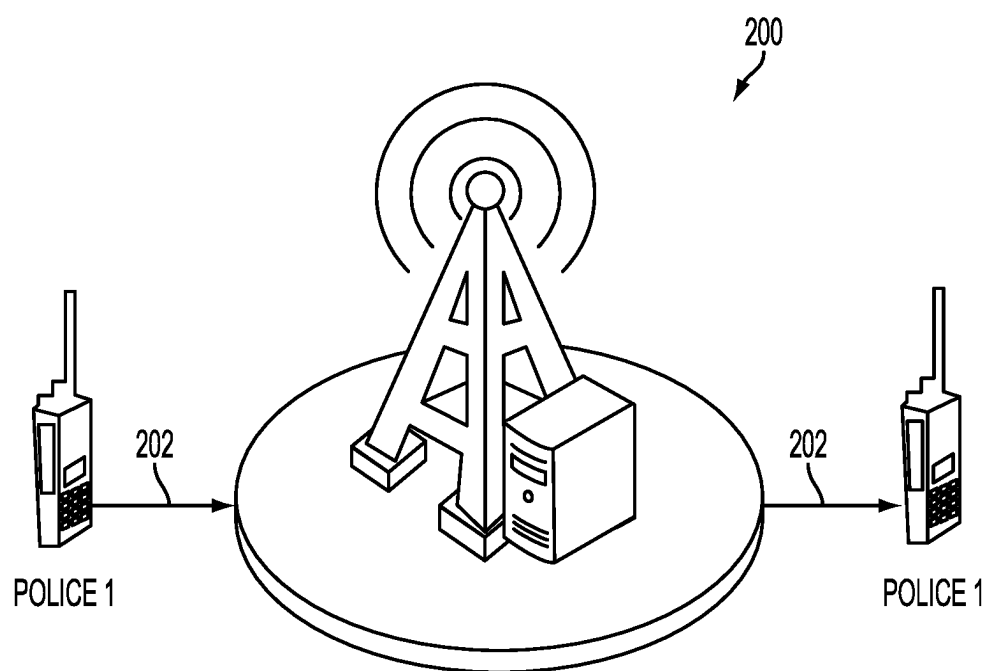


FIG. 2

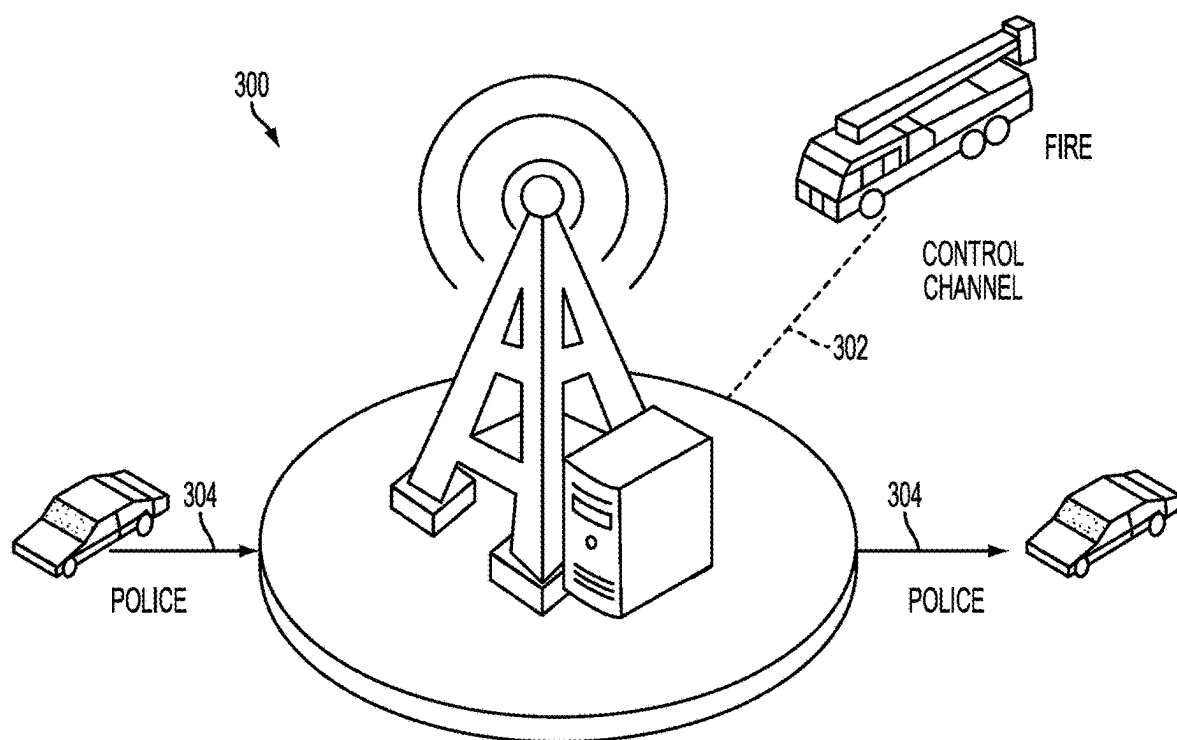


FIG. 3

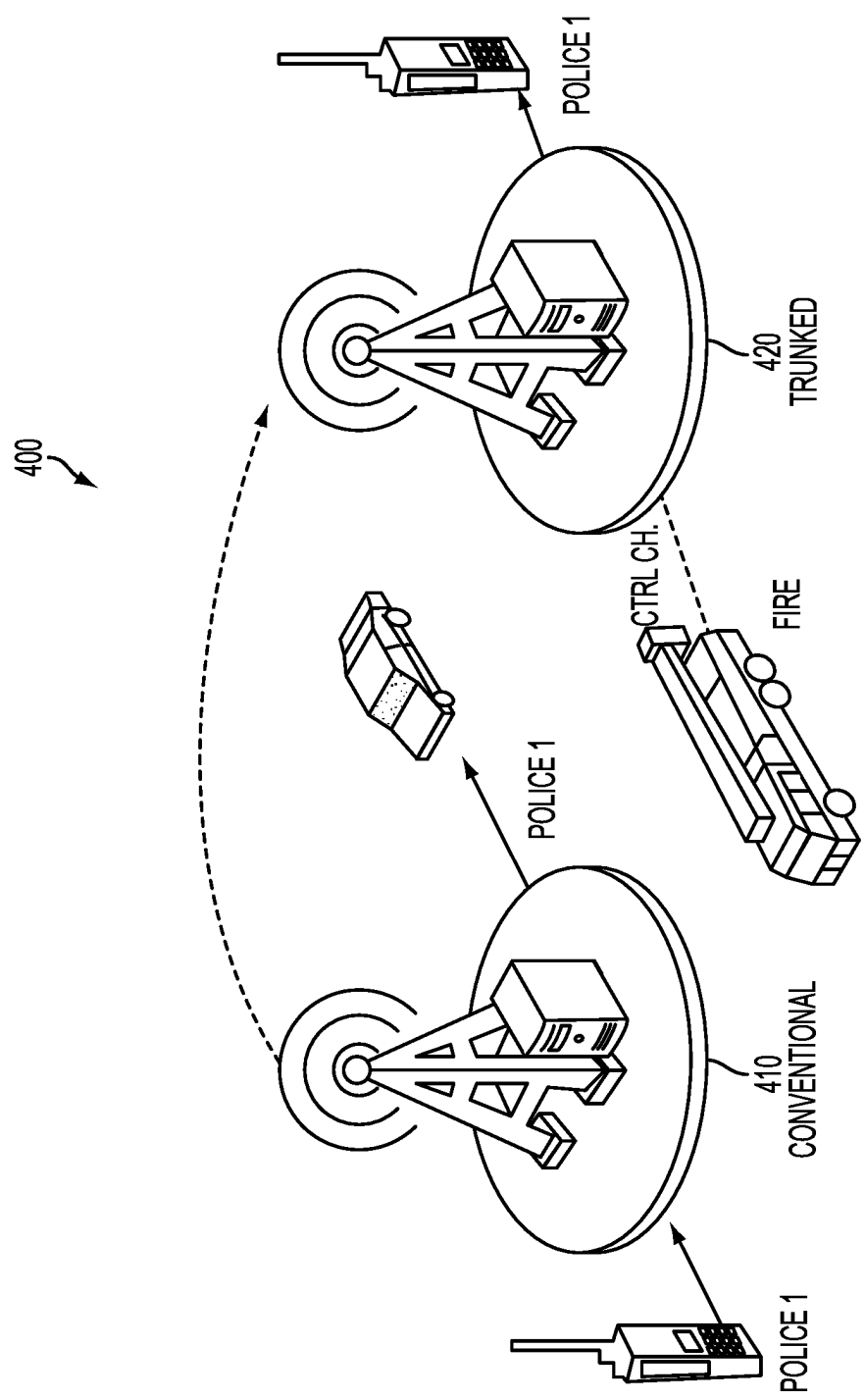


FIG. 4

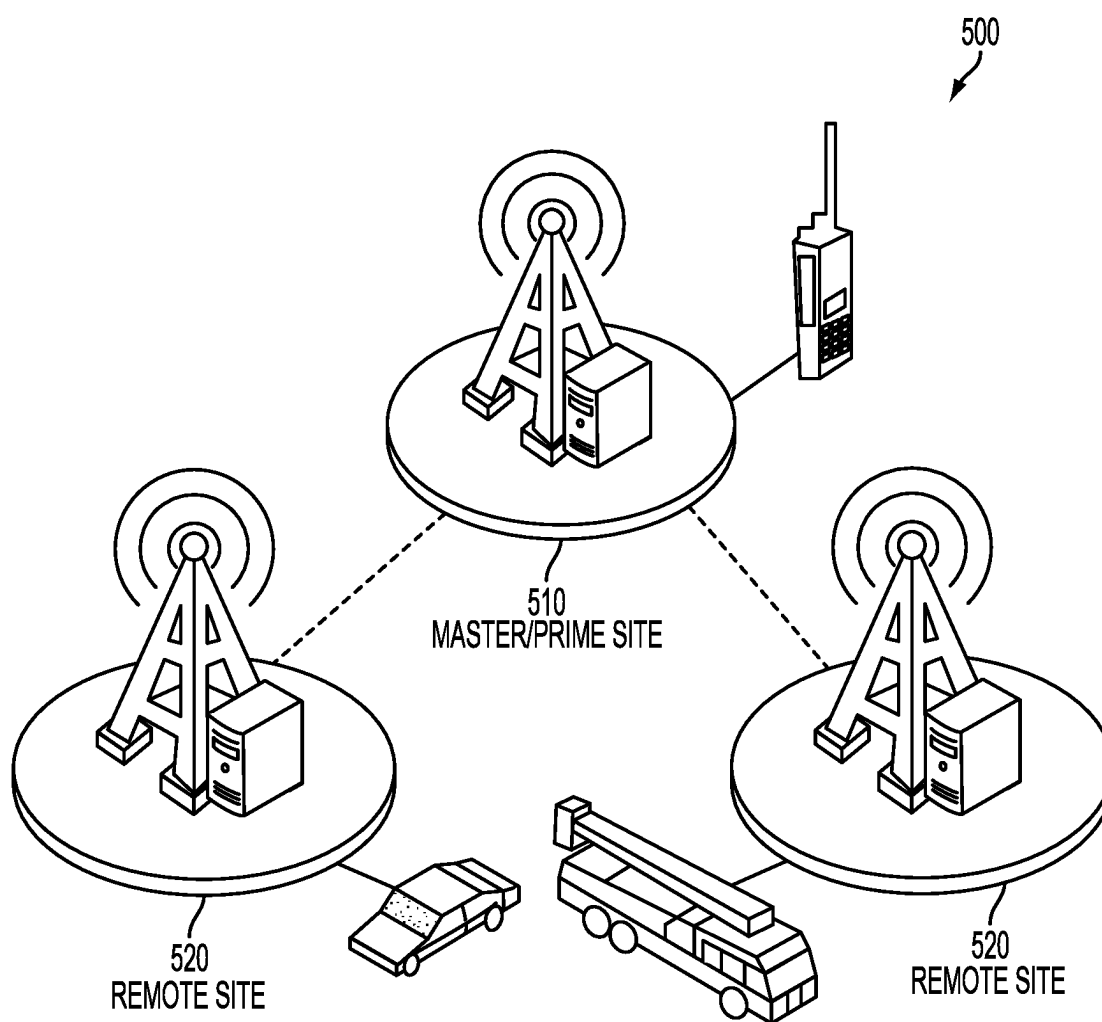


FIG. 5

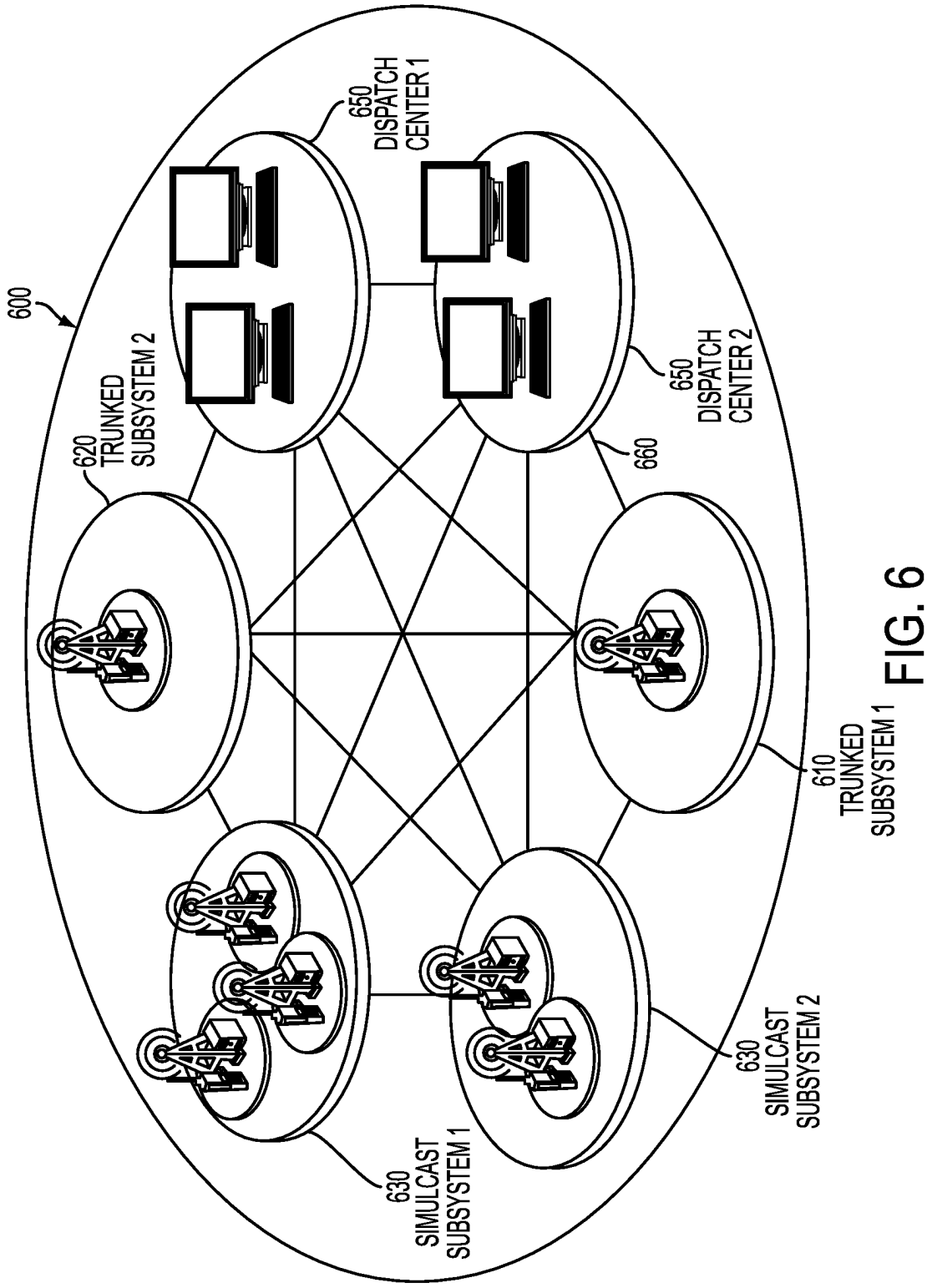


FIG. 6

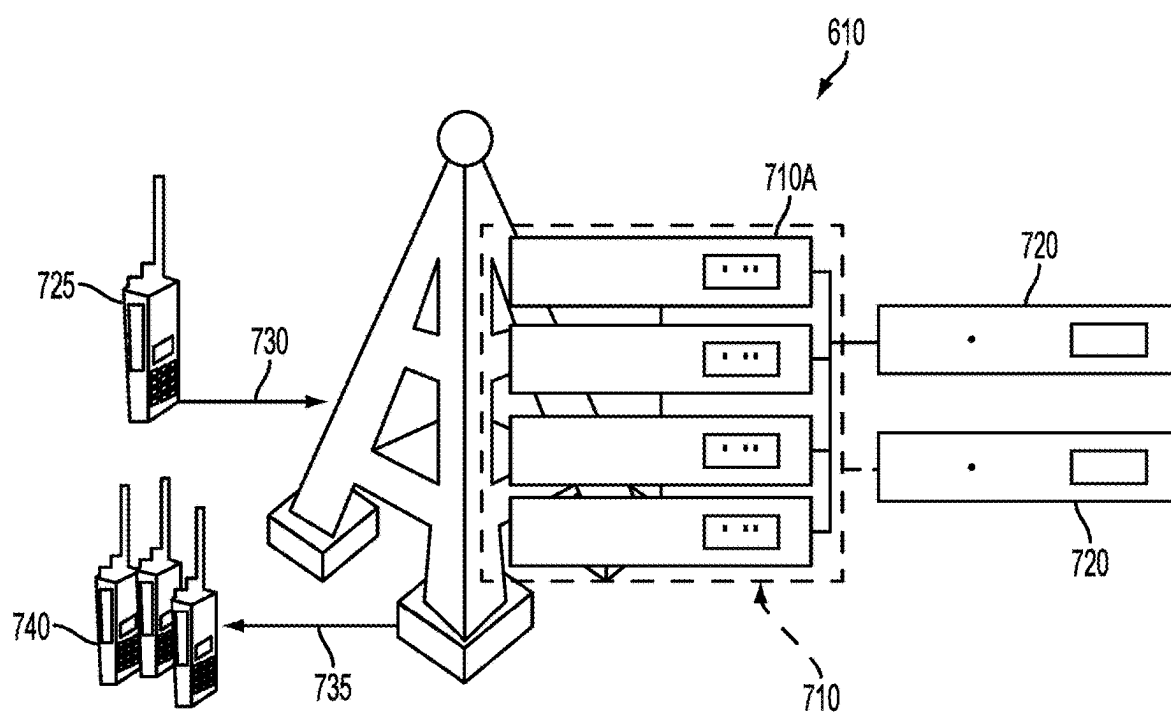


FIG. 7

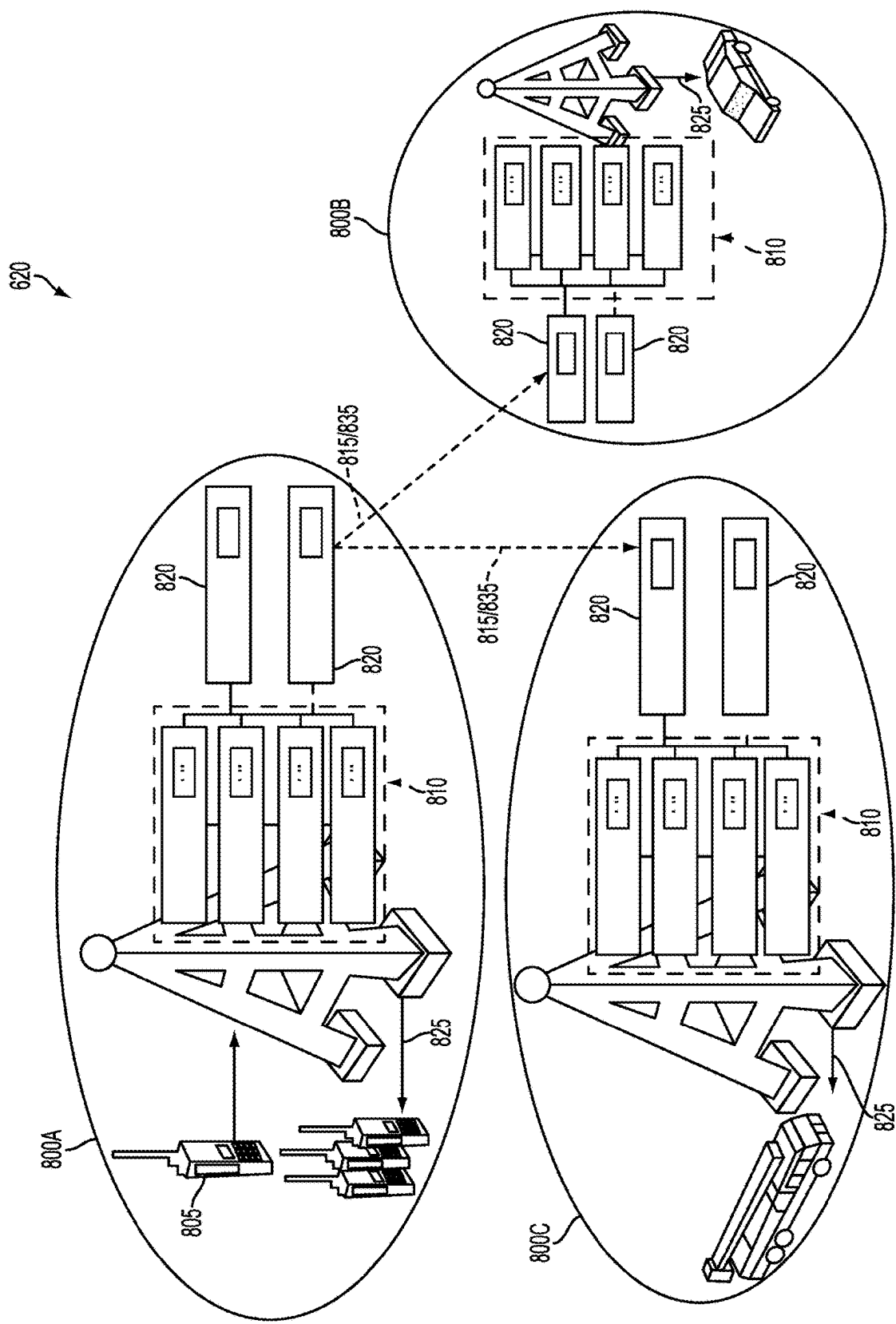
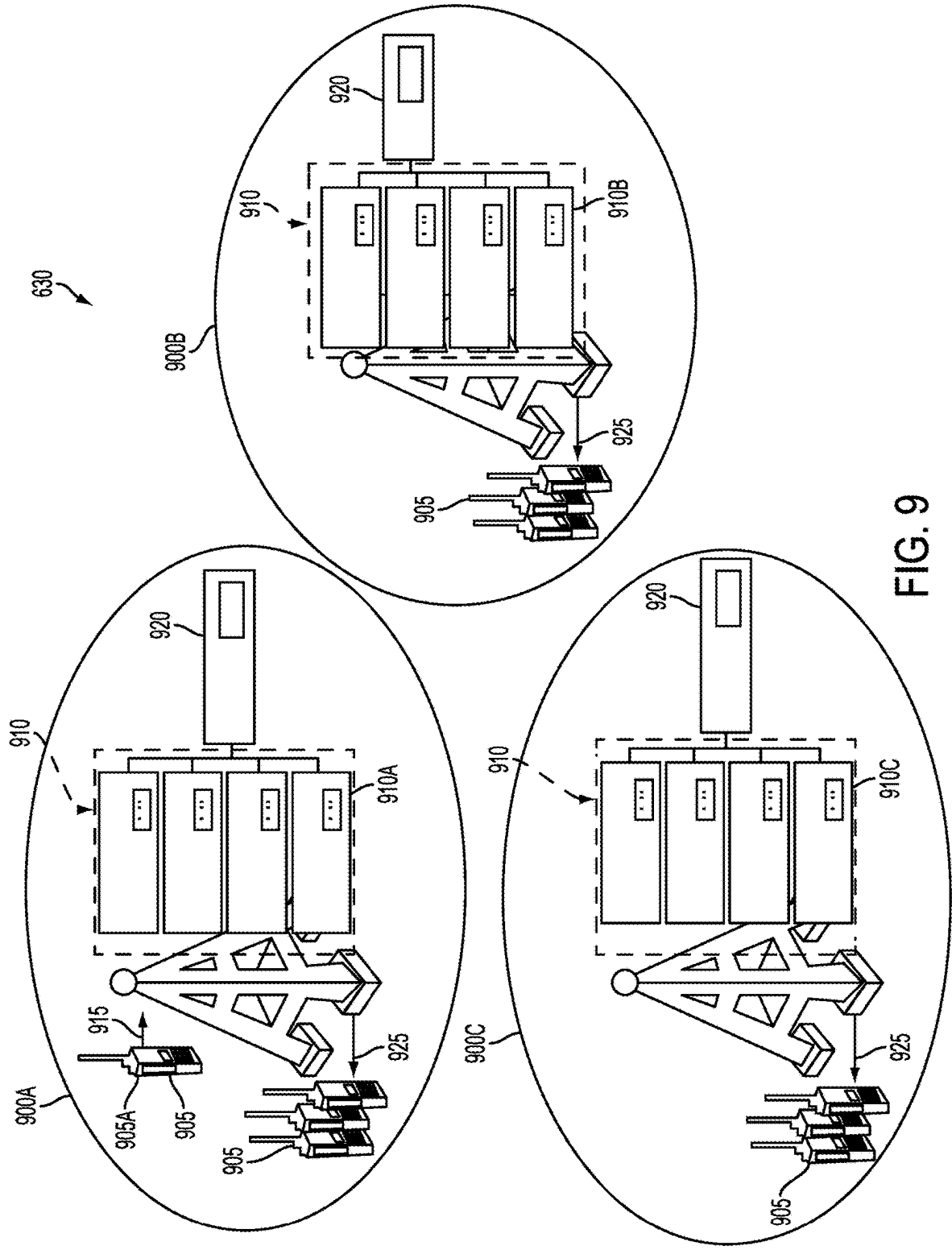


FIG. 8



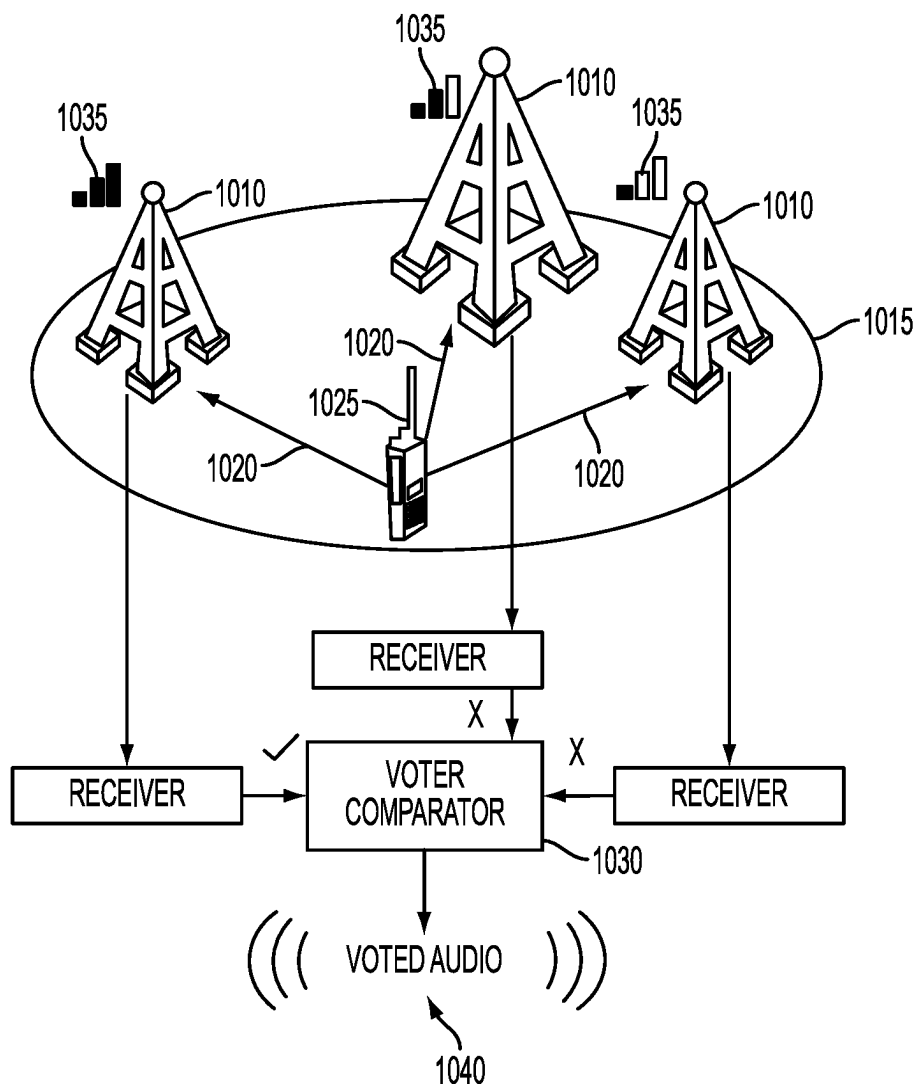


FIG. 10

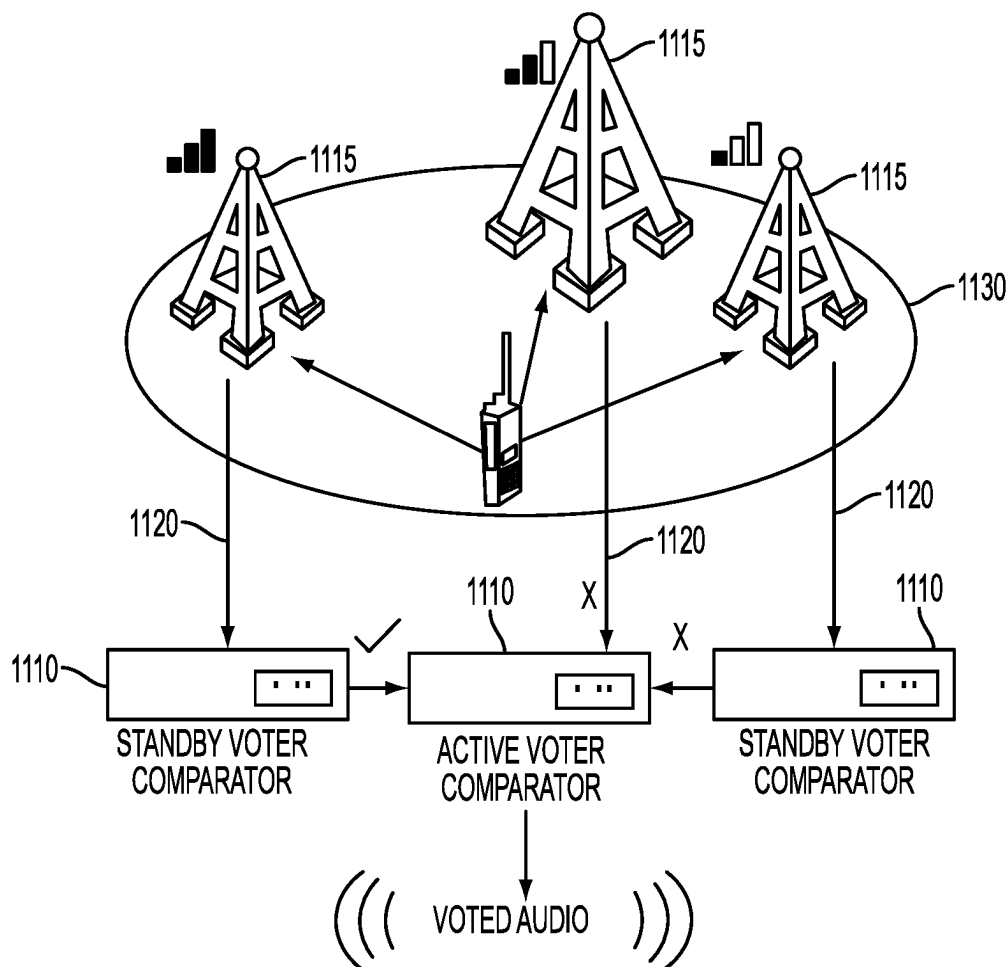


FIG. 11

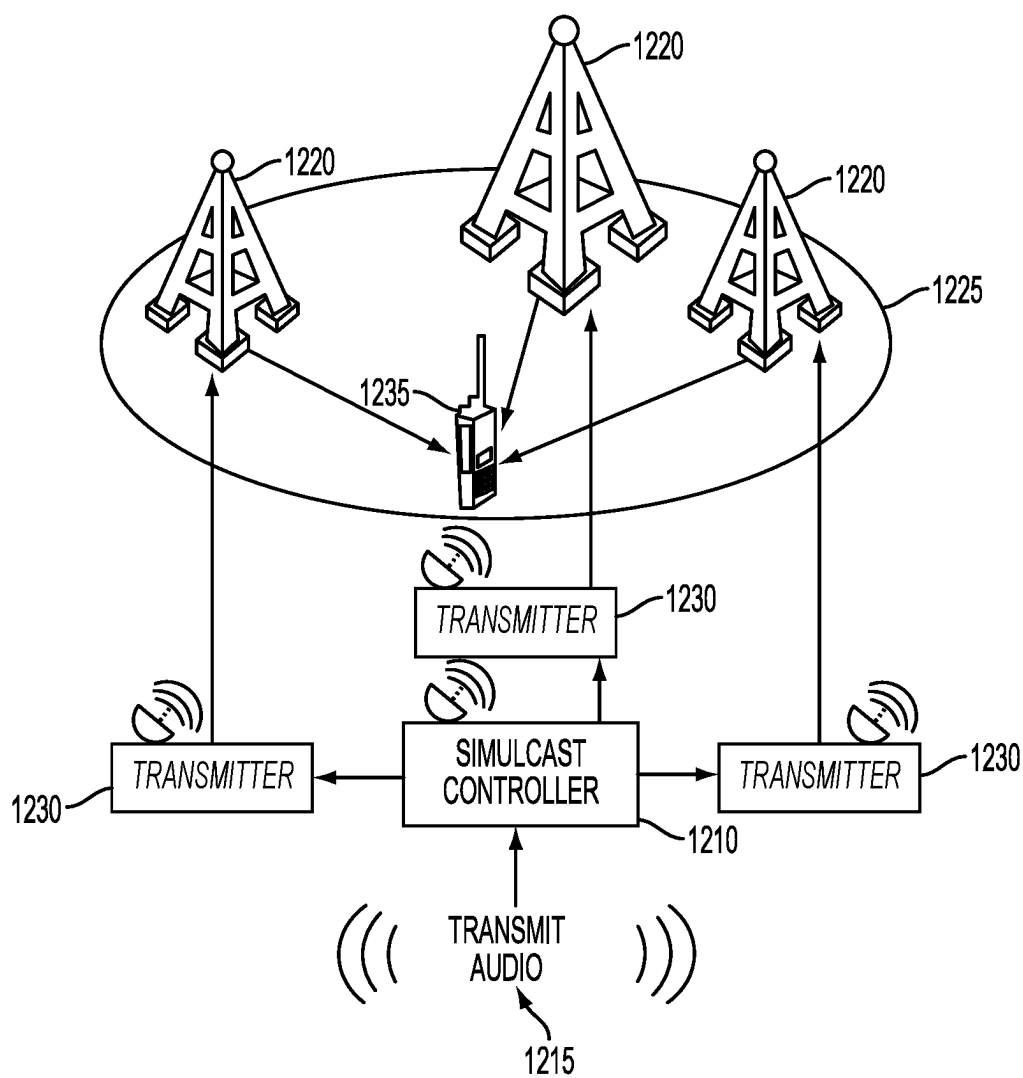


FIG. 12

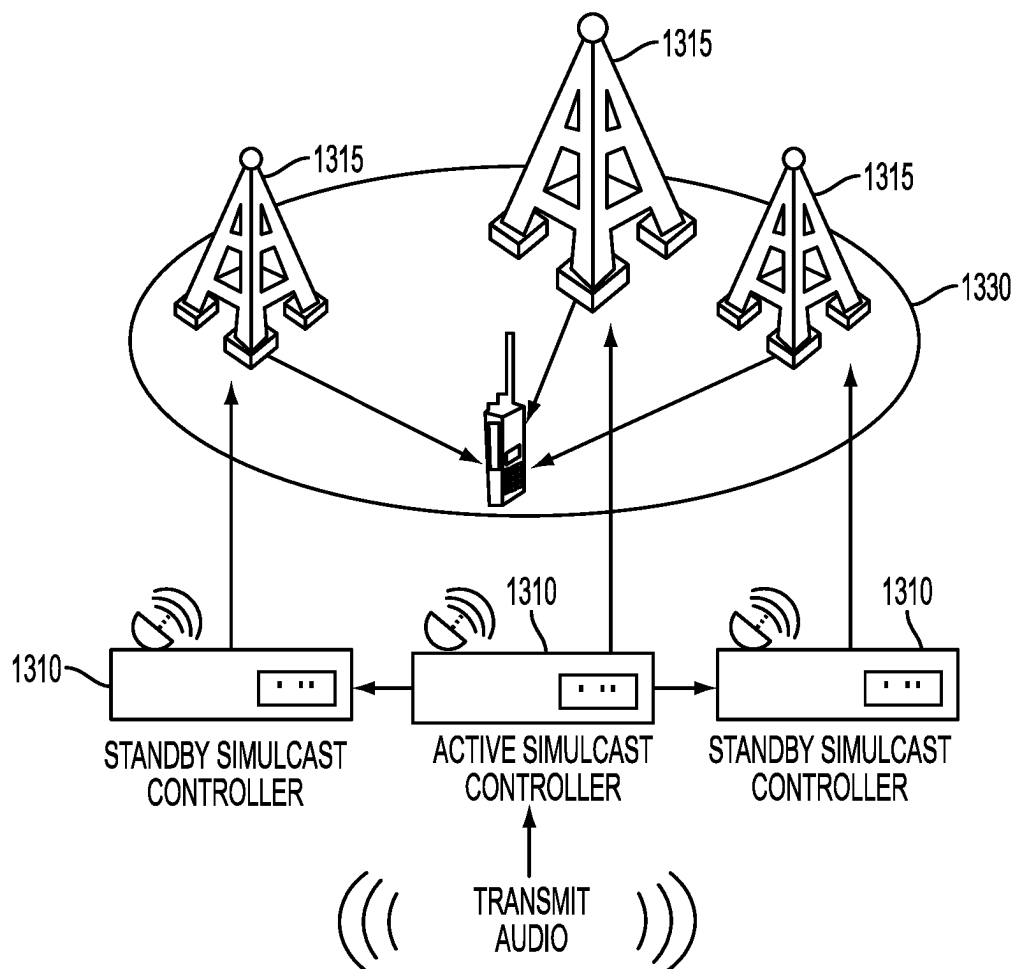


FIG. 13

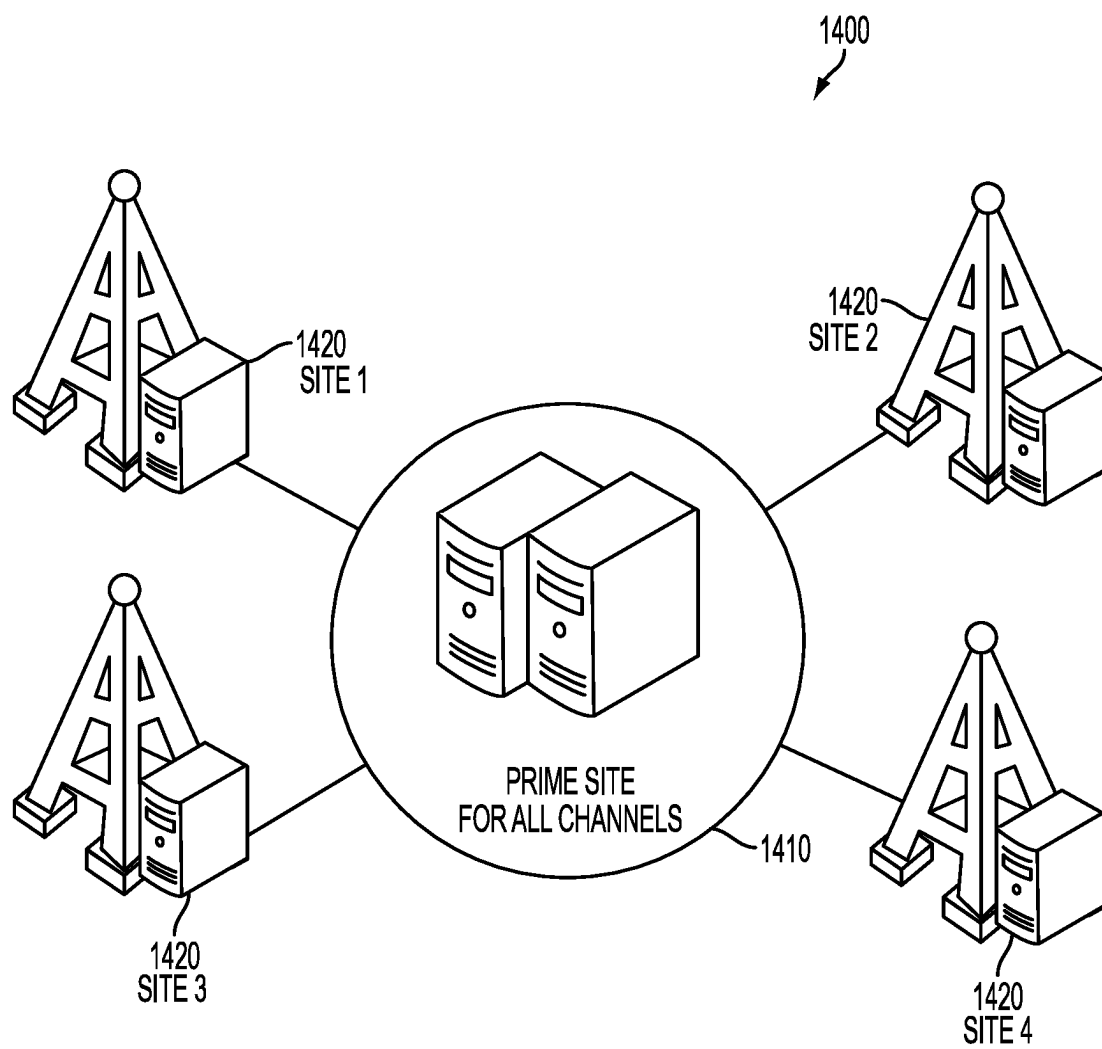


FIG. 14

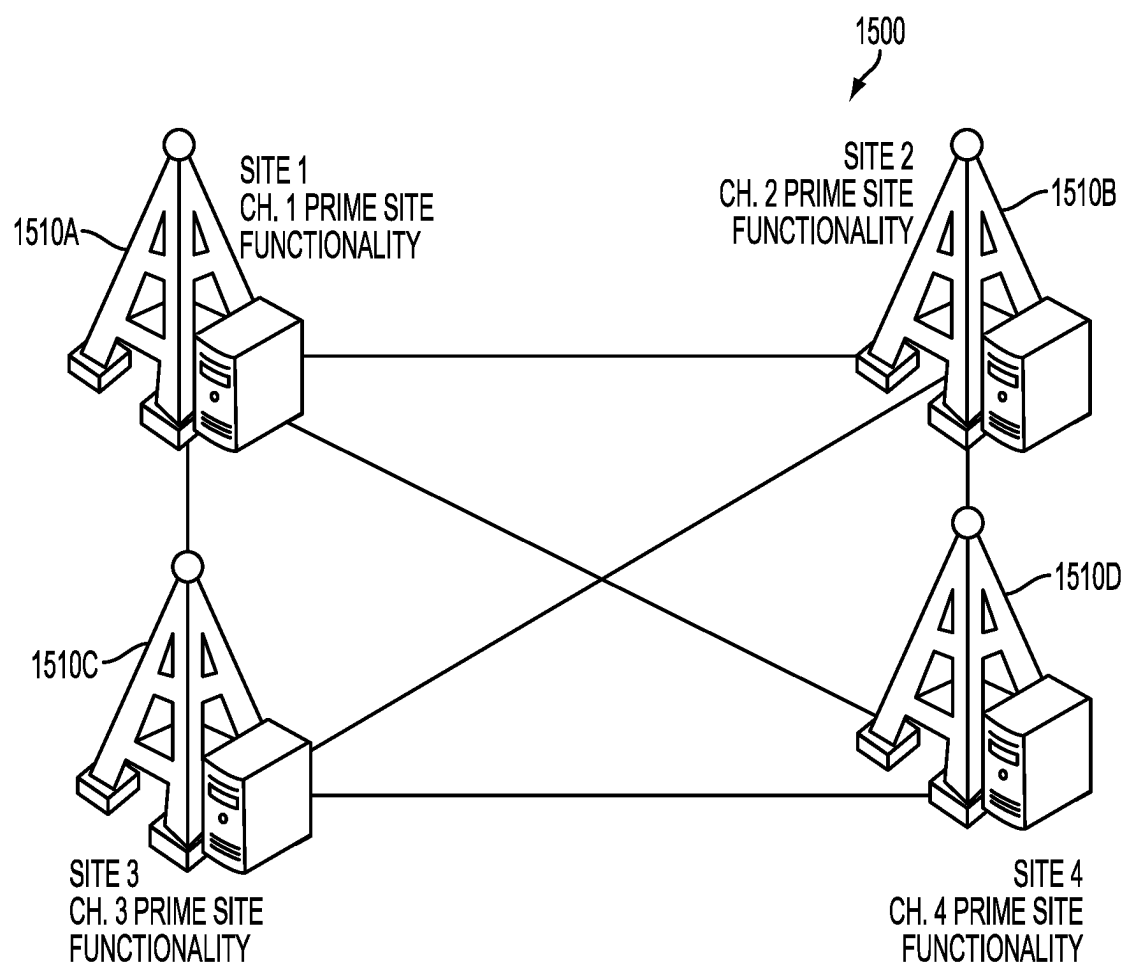


FIG. 15

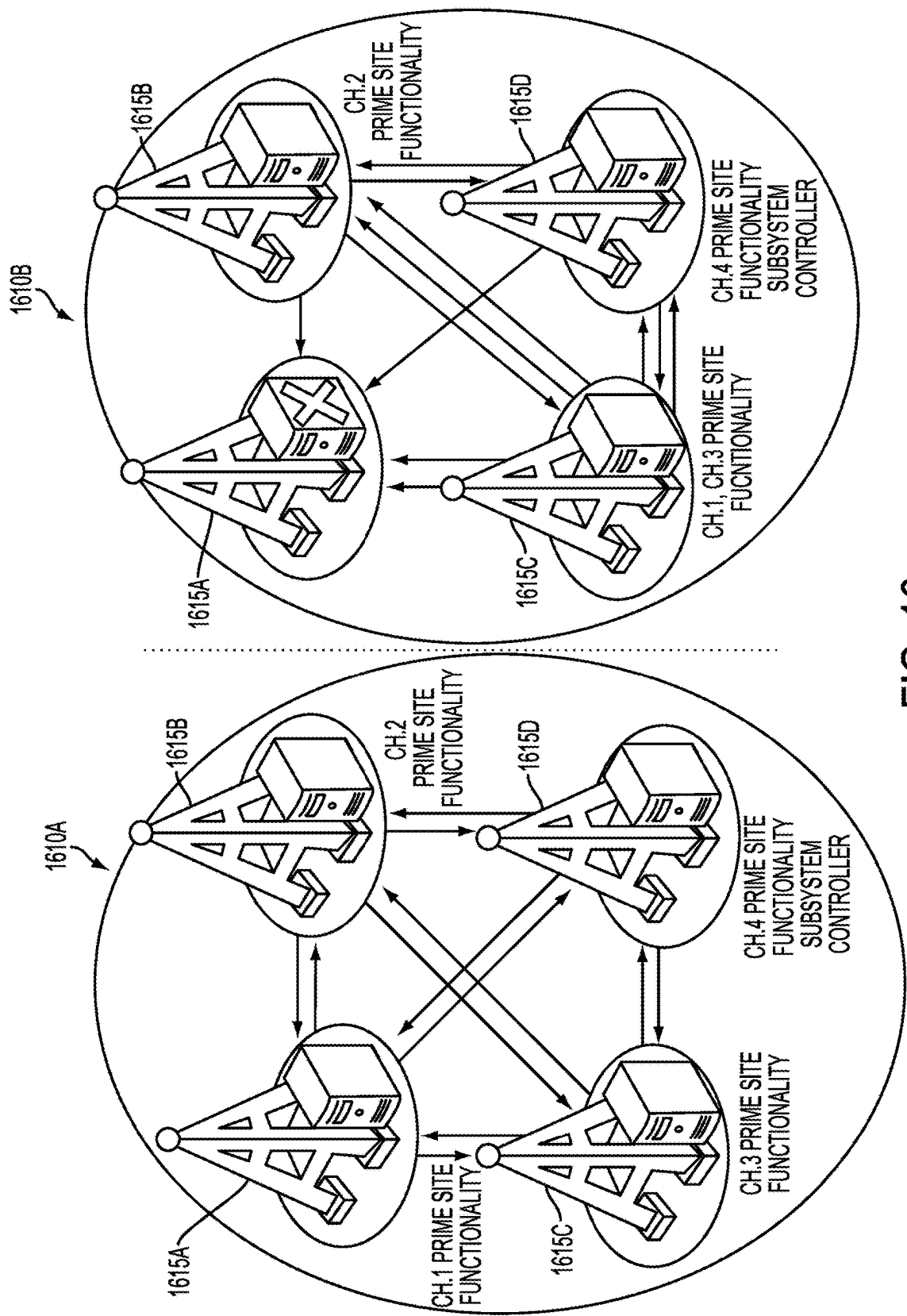


FIG. 16

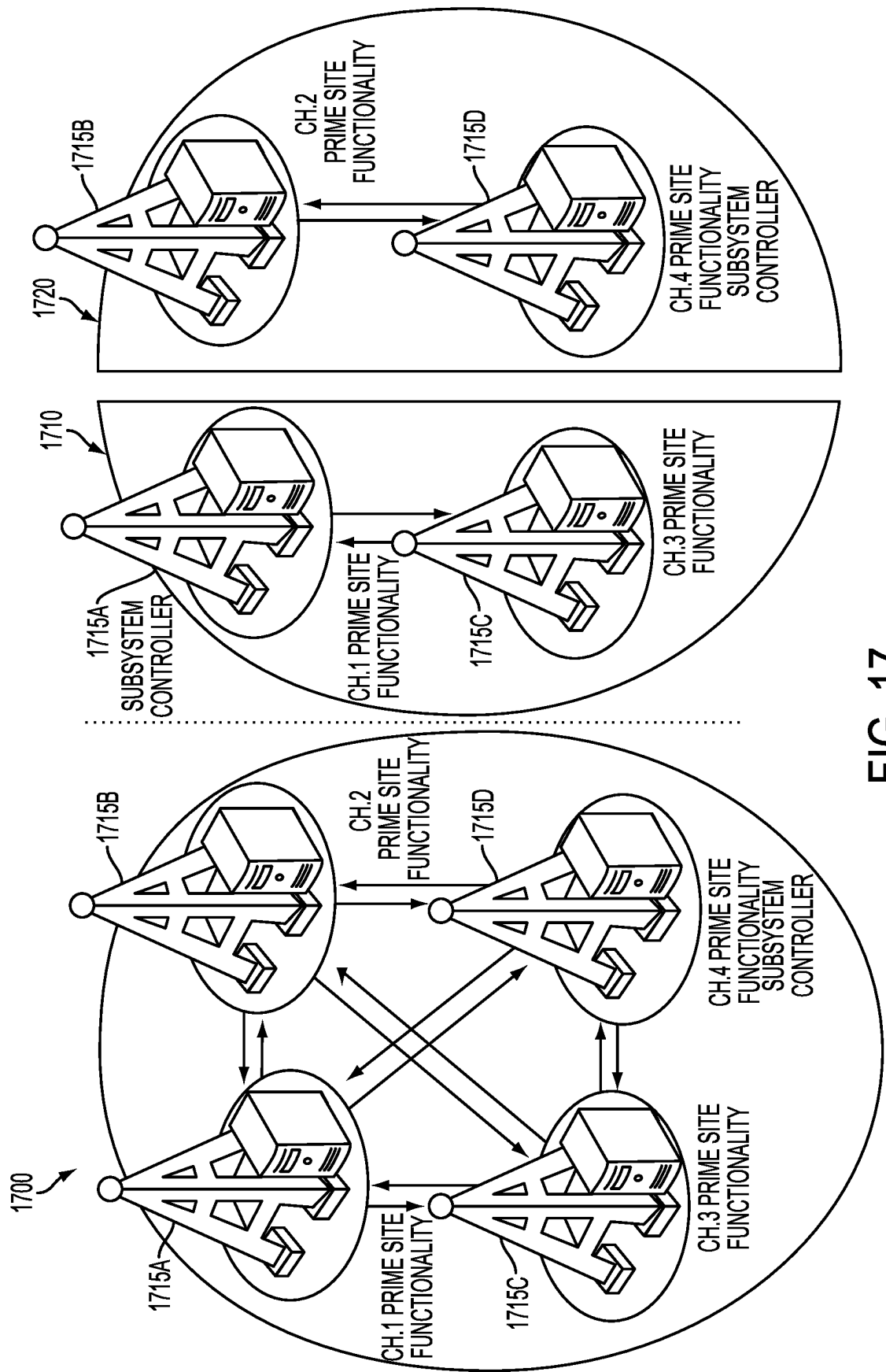


FIG. 17

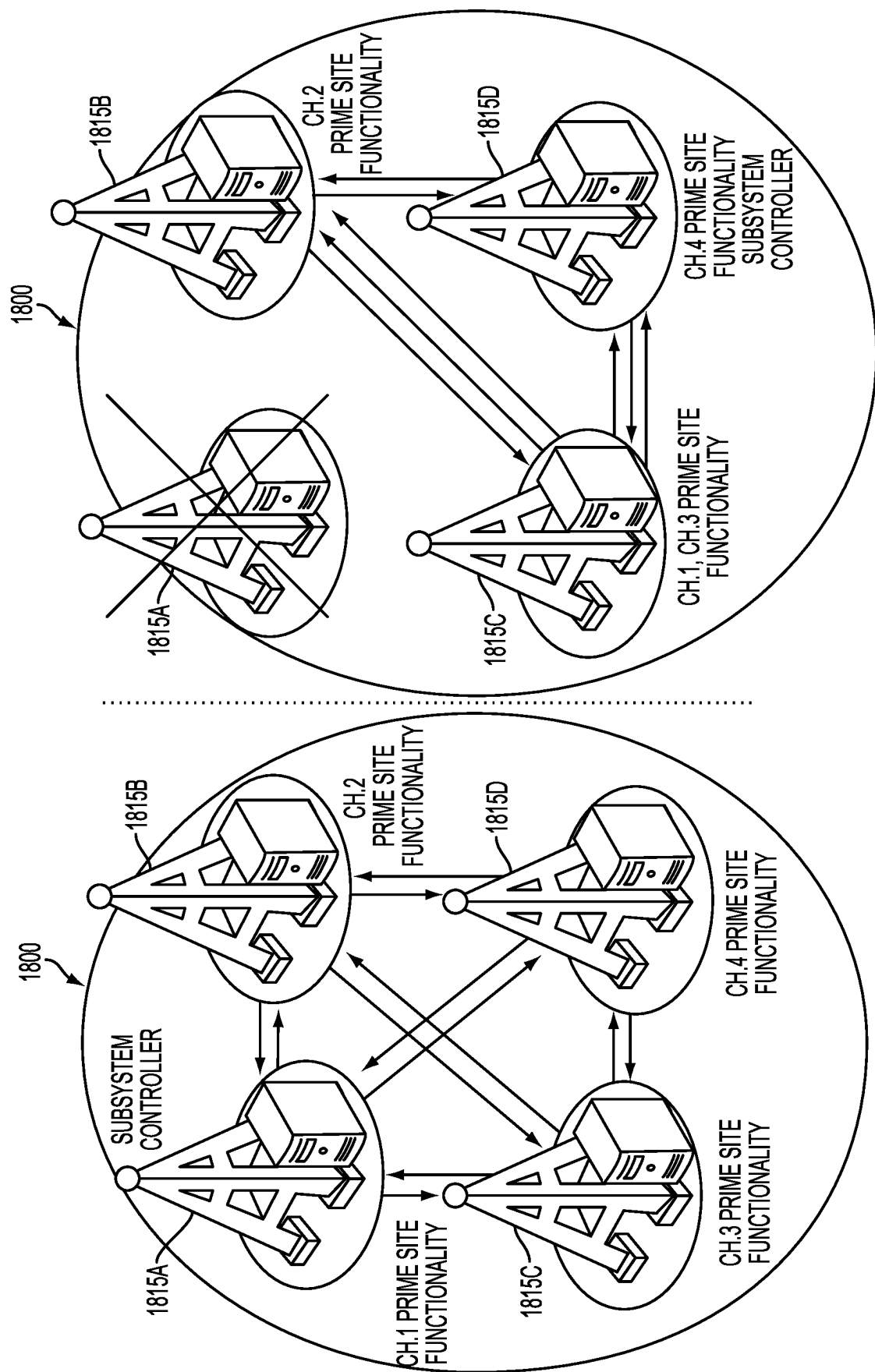


FIG. 18

DISTRIBUTED LAND MOBILE RADIO ARCHITECTURES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/595,143, filed Mar. 4, 2024, which is a continuation of U.S. patent application Ser. No. 17/981,070, filed Nov. 4, 2022, which is a continuation of U.S. patent application Ser. No. 17/100,213, filed Nov. 20, 2020, which is a continuation of U.S. patent application Ser. No. 16/543,302, filed Aug. 16, 2019, which is a continuation of U.S. patent application Ser. No. 15/494,391, filed Apr. 21, 2017, which is a continuation of U.S. patent application Ser. No. 14/217,150, filed Mar. 17, 2014, which claims the benefit of U.S. Provisional Patent Application No. 61/790,588, filed Mar. 15, 2013, all of which are incorporated herein by reference in their entireties and for all purposes.

BACKGROUND

[0002] The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

[0003] Land Mobile Radio (LMR) systems are deployed by organizations requiring instant communication between geographically dispersed and mobile personnel. Current LMR systems can be configured to provide for radio communications between one or more sites and subscriber radio units in the field. A subscriber radio unit (hereinafter “radio”) may be a mobile unit or a portable unit. LMR systems can be as simple as two radio units communicating between themselves over preset channels, or they can be complex systems that include hundreds of radio units and multiple sites. Typical users of LMR systems include police departments, fire departments, medical personnel, security personnel, EMS, and the military.

[0004] Conventional and trunking LMR system architecture may include multiple LMR sites, each utilizing various equipment including, for example, dedicated site controllers, repeaters, voter comparator and simulcast controllers. Specifically, in simulcast system architecture, a prime site is deployed that hosts the site controllers, simulcast controllers and voter comparators. As the LMR system expands, additional equipment is needed, which becomes increasingly expensive to provide and maintain. Furthermore, each site in the LMR system is often controlled by equipment located at one of the sites comprising the LMR system or by the equipment located at the prime site. Accordingly, when such equipment fails, corresponding portions of the LMR system also fail. As such, conventional and trunking LMR system architecture lacks redundancy and, therefore, is often subject to single points of failure, thereby compromising the integrity of the LMR system architecture.

SUMMARY

[0005] In one embodiment, the present disclosure provides a system for providing communication in a distributed LMR system architecture, the distributed LMR system architecture comprising one or more subsystems in communication with a data network, the system comprising: one or more LMR sites comprising at least one of the one or more subsystems; one or more subsystem controllers disposed at each of the one or more LMR sites comprising the at least

one subsystem, each subsystem controller having at least an active mode and a standby mode, wherein at least one subsystem controller is operable in the active mode to control communication between the one or more LMR sites in the at least one subsystem; and one or more repeaters disposed at each of the plurality of sites in the at least one subsystem, each of the repeaters operable to provide a communication channel, wherein each repeater has at least an active mode and a standby mode, and wherein at least one repeater is operable in the active mode to perform at least one of a simulcast controller operation and a voter comparator operation.

[0006] In another embodiment, the present disclosure provides a method for providing communication in a distributed land mobile radio (LMR) system architecture, the distributed LMR system architecture comprising one or more subsystems in communication with a data network, the method comprising: providing a subsystem controller in each of a plurality of LMR sites comprising one of the subsystems, each subsystem controller having at least an active mode and a standby mode; operating one of the subsystem controllers in the active mode to control communication between the plurality of LMR sites; operating the remaining subsystem controllers in the standby mode; providing a plurality of repeaters at each of the plurality of LMR sites comprising the subsystem, each repeater having at least an active mode and a standby mode; operating at least one of the repeaters in the active mode to perform at least one of a simulcast controller operation and a voter comparator operation; and operating the remaining repeaters in the standby mode.

[0007] Further embodiments and apparatuses, including other areas of applicability, will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure in any manner.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] For a more complete understanding of various embodiments of the present invention and the advantages thereof, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts, and in which:

[0009] FIG. 1 illustrates an example embodiment of a centralized LMR architecture;

[0010] FIG. 2 illustrates an example embodiment of a conventional LMR system;

[0011] FIG. 3 illustrates an example embodiment of a trunked LMR system;

[0012] FIG. 4 illustrates an example embodiment of a hybrid LMR system;

[0013] FIG. 5 illustrates an example embodiment of a simulcast LMR system;

[0014] FIG. 6 illustrates an example embodiment of an LMR system incorporating a distributed architecture;

[0015] FIG. 7 illustrates an example embodiment of the first trunked LMR subsystem provided in FIG. 6;

[0016] FIG. 8 illustrates an example embodiment of the second trunked LMR subsystem provided in the distributed architecture of FIG. 6;

[0017] FIG. 9 illustrates an example embodiment of one of the simulcast LMR subsystems provided in the distributed architecture of FIG. 6;

[0018] FIG. 10 illustrates an example embodiment of system for providing receiver voting in a centralized architecture;

[0019] FIG. 11 illustrates an example embodiment of a system for providing receiver voting in a distributed architecture;

[0020] FIG. 12 illustrates an example embodiment of system for providing simulcast communication in a centralized architecture;

[0021] FIG. 13 illustrates an example embodiment of a system for providing simulcast communication in a distributed architecture;

[0022] FIG. 14 illustrates an example embodiment of a centralized simulcast subsystem;

[0023] FIG. 15 illustrates an example embodiment of a trunked simulcast subsystem in a distributed architecture;

[0024] FIG. 16 illustrates an example embodiment demonstrating simulcast controller and voter comparator redundancy in a distributed simulcast LMR architecture;

[0025] FIG. 17 illustrates an example embodiment demonstrating network failure redundancy in a distributed simulcast LMR architecture; and

[0026] FIG. 18 illustrates an example embodiment demonstrating site redundancy in a distributed simulcast LMR architecture.

DETAILED DESCRIPTION OF THE DRAWINGS

[0027] In the following detailed description and accompanying drawings, numerous specific details are set forth to provide a thorough understanding of the present disclosure. However, those skilled in the art will appreciate that the present disclosure may be practiced, in some instances, without such specific details. In other instances, well-known elements have been illustrated in schematic or block diagram form in order not to obscure the present disclosure in unnecessary detail. Additionally, for the most part, specific details, and the like, have been omitted inasmuch as such details are not considered necessary to obtain a complete understanding of the present disclosure, and are considered to be within the purview of persons of ordinary skill in the relevant art.

[0028] It is further noted that, unless indicated otherwise, all functions described herein may be performed in hardware or as software instructions for enabling a computer, radio, or other device to perform predetermined operations, where the software instructions are embodied on a computer readable storage medium, such as RAM, a hard drive, flash memory, or other type of computer readable storage medium known to a person of ordinary skill in the art. In certain embodiments, the predetermined operations of the computer, radio, or other device are performed by a processor such as a computer or an electronic data processor in accordance with code such as computer program code, software, firmware, and, in some embodiments, integrated circuitry that is coded to perform such functions. Furthermore, it should be understood that various operations described herein as being performed by a user may be operations manually performed by the user, or may be automated processes performed either with or without instruction provided by the user.

[0029] An LMR system may employ a centralized architecture whereby various LMR subsystems are connected by

a central network controller and associated network equipment. FIG. 1 illustrates an example of a centralized LMR architecture 100, which includes various LMR subsystems 110, dispatch stations 115, and gateway equipment 120 connected by a central network controller 125. The central network controller 125 includes equipment for operating and controlling each of the various LMR subsystems 110, dispatch stations 115, and gateway equipment 120. The various LMR subsystems 110 may include any of a conventional LMR system, trunking LMR system, hybrid LMR system, or wide area systems such as a simulcast LMR system or multicast LMR system. Examples of such LMR systems are briefly discussed below with reference to FIGS. 2-5.

[0030] FIG. 2 illustrates an example of a conventional LMR system 200. Conventional systems are typically deployed in regions covering large geographic areas and/or comprising a moderate quantity of users. In a conventional system, a dedicated repeater channel is provided for system user groups, and the user chooses the channel or channels on which he wishes to communicate. Often, system user groups are organized based on responsibility, such as Fire, Police, EMS, Public Works, and Mutual Aid. In most cases, each of these groups has a corresponding dedicated repeater channel 202, which provides a frequency for communicating a call.

[0031] FIG. 3 illustrates an example of a trunked LMR system 300. Trunked systems are typically deployed in regions covering moderate geographical areas and/or comprising a large quantity of users. Trunked systems have a shared pool of repeater channels for use among system user groups. The repeater channels at each site are divided into a control channel 302 and multiple voice channels 304. The control channel 302 registers a radio into the system and dynamically coordinates radio talkgroup Push-to-Talk (PTT) with an available voice channel.

[0032] In a trunked radio system, system talkgroups are often organized based on responsibility, such as Fire, Police, EMS, Public Works, and Mutual Aid. The user selects the talkgroup with which he wishes to communicate, and the trunked system then allocates the radio channel used for the voice transmission. For LMR systems having multiple groups with access to multiple channels at each site, a trunked system may be implemented to increase the system's efficiency.

[0033] A hybrid system combines conventional and trunked repeater channels into a single system. In hybrid systems, users can be organized functionally for either the conventional or trunked part of the system, as needed. FIG. 4 illustrates an example embodiment of a hybrid LMR system 400, which includes a conventional site 410 and a trunked site 420.

[0034] In addition to the foregoing, LMR system types may include wide area systems, which, in some embodiments, are designed to enable radios to move throughout an area without their users needing to change channels while roaming. A simulcast system is an example of a wide area system. An example of a simulcast system 500 is illustrated in FIG. 5. The simulcast system 500 includes a single master site 510 (also referred to herein as a prime site) and multiple radio sites 520. The master site 510 synchronizes the system timing so that calls are transmitted simultaneously to all sites for a given repeater channel. Thus, a call is transmitted simultaneously to all sites at the same frequency. This synchronization reduces the quantity of frequencies needed for the system and simplifies frequency coordination.

[0035] Another example of a wide area LMR system is a multicast system. In multicast systems, different transmitters within adjacent geographic areas communicate on different radio channel frequencies. The multicast system switches the user to the proper channel automatically. The multicast system configuration offers similar coverage advantages of a simulcast system at a reduced cost. However, multicast systems require multiple frequencies, and their users need to change mobile channels as they move between sites.

[0036] Referring again to FIG. 1, the central network controller 125 includes equipment for operating and controlling each of the various LMR subsystems 110, dispatch stations 115, and gateway equipment 120. As the LMR system 100 expands, additional capacity is needed in the network controller often requiring additional network controller equipment, which becomes increasingly expensive to provide and maintain. Furthermore, sites in the subsystems 110 may be controlled by equipment located at the central network controller 125. When such equipment fails, the LMR system 100 fails. As such, centralized LMR architecture lacks redundancy and, therefore, is often subject to single points of failure, thereby compromising the integrity of the LMR system architecture.

[0037] The present disclosure provides a system and method for providing communication in a distributed LMR system architecture. The distributed architecture eliminates the need for a central network controller and associated network equipment. Instead, the functionality of the network controller is distributed among controllers at each of the subsystems comprising the LMR system, thereby providing peer-to-peer communication over an internet protocol (IP) network.

[0038] FIG. 6 illustrates an example of an LMR system 600 incorporating a distributed architecture in accordance with the present disclosure. The distributed LMR architecture 600 provided in FIG. 6 is comprised of various LMR subsystems 610-630 and dispatch stations 650 connected via IP connections 600 comprising an IP network. In the embodiment illustrated in FIG. 6, the distributed LMR architecture 600 incorporates first and second trunked subsystems 610 and 620 and simulcast subsystems 630, however, it should be appreciated that other LMR subsystems may be included, such as, for example, conventional, hybrid, multicast, or any other LMR systems discussed herein.

[0039] As mentioned above, the distributed LMR architecture 600 eliminates the central network controller and associated equipment that is typically provided with a centralized architecture, and instead distributes the functionality of the central network controller and associated equipment among subsystem controllers deployed at each of the subsystems 610-630 comprising the distributed LMR architecture 600. In some embodiments, the central network controller functionality and associated equipment may also be distributed among dispatch stations 650.

[0040] As discussed in greater detail below, the distributed LMR architecture disclosed herein incorporates repeaters, subsystem controllers, network management systems, and dispatch consoles. In some embodiments, these components are IP-based and may be managed remotely over the IP network.

[0041] Repeaters provide channels/frequencies for over-the-air communication and, in some embodiments, are equipped with circuitry to provide integrated voter comparator and simulcast controller functionality/operations.

[0042] Subsystem controllers provide interface and gateway functionality for interconnecting multiple types of LMR subsystems through a common IP network. The subsystem controllers enable dispatch console control of local repeaters, provide distributed call control and mobility management functions, and enable direct routing of calls between conventional and trunked systems and/or dispatch consoles without talkgroup patching. The distributed architecture of the disclosed system enables each subsystem controller to perform central network controller functionality for a call originating from its local subsystem, thereby eliminating the need for a dedicated central network controller. As discussed in greater detail below, providing a subsystem controller at each site in a subsystem provides multi-level redundancy of the controller functionality, and allows for communication in case of equipment or site failure.

[0043] Network management systems provide redundant, web-based, and centralized network management functionality for the infrastructure comprising the distributed architecture system, including the various LMR subsystems (e.g., conventional, trunked, etc.), subsystem controllers, and dispatch consoles. The network management systems provide management and deployment of subscriber and talkgroup records; radio administration including radio inhibit, dynamic regrouping, and radio check; agency specific management of subscriber records, talkgroup records, and reporting; and pre-defined and custom roles that restrict operator access and activity based on access credentials. The network management systems also provide real-time fault monitoring of system components, extensive reports covering system usage and user activities, real-time monitoring of user and channel activities, and full redundancy capability.

[0044] Dispatch consoles provide interoperability via direct IP connection to the LMR subsystems. In some embodiments, the dispatch consoles are IP-based and fully distributed with no requirement for central control equipment, thereby allowing extensive scalability and expansion with no single point of failure.

[0045] Reference is now made to FIG. 7, which illustrates an example embodiment of the first trunked LMR subsystem 610 of FIG. 6. In the example embodiment illustrated in FIG. 7, the first trunked subsystem 610 comprises a single site trunked subsystem, however, it should be appreciated that the trunked subsystem could include multiple sites. The trunked subsystem 610 includes a plurality of repeaters 710 and two subsystem controllers 720. Each of the repeaters 710 represents a channel (a combination of transmit and receive operations), wherein one of the repeaters (e.g., 710A) operates a control channel 730, while the remaining repeaters 710 are designated for voice operation.

[0046] The trunked subsystem 610 provides redundancy by incorporating two subsystem controllers 720. One of the subsystem controllers 720 is active, and the other is on standby. If the active subsystem controller 720 fails, then the standby subsystem controller 720 becomes active to provide a fail-safe transition with no visible impact to the radio users 725 and 740. As discussed above, the local subsystem controller 720 performs call controls, thereby eliminating the need for a central controller.

[0047] In accordance with the embodiment illustrated in FIG. 7, the single site trunked subsystem 610 operates in accordance with the following example for hosting a call. A radio user 725 initiates a call through the control channel 730. The active subsystem controller 720 processes the

user-initiated call and assigns a voice channel 735 for voice communication. The radio users 725 and 740 communicate with each other over the assigned voice channel 735.

[0048] Reference is now made to FIG. 8, which illustrates an example embodiment of the second trunked LMR subsystem 620 of FIG. 6. In the example embodiment illustrated in FIG. 8, the second trunked subsystem 620 comprises a wide area multicast trunked subsystem. The multicast trunked subsystem 620 comprises a plurality of sites 800A-800C connected with each other to form a wide area trunked system. Each of the sites 800A-800C is similar to the single trunked site discussed above with respect to FIG. 7 and, therefore, includes a plurality of repeaters 810 and two subsystem controllers 820. In some embodiments, adjacent sites in the multicast trunked subsystem 620 operate at a different frequency to ensure there is no radiofrequency (RF) interference between the sites. However, in some embodiments, frequencies may be reused in non-overlapping sites.

[0049] The sites 800A-800C are connected via their respective subsystem controllers 820, thereby eliminating the need for a central controller. The subsystem controllers 820 communicate directly with each other to setup a wide area call between interested sites. For example, a call originating from a first site (e.g., site 800A) is transferred to other interested sites (e.g., sites 800B and 800C) using the local subsystem controllers 820.

[0050] In accordance with the embodiment illustrated in FIG. 8, the multicast trunked subsystem 620 operates in accordance with the following example for hosting a call. A radio user 805 from site 800A originates a call. The active subsystem controller 820 in site 800A determines the other sites interested in the call and sends them a call notification 815. The active subsystem controller 820 located in each site 800A-800C assigns a voice channel 825 to the call. The active subsystem controller 820 in site 800A transfers the call 835 to the other subsystem controllers 820, and the radios use the assigned voice channel 825 to transmit or receive within their respective site.

[0051] Reference is now made to FIG. 9, which illustrates an example embodiment of one of the simulcast LMR subsystems 630 of FIG. 6. In the example embodiment illustrated in FIG. 9, the simulcast subsystem 630 is a trunked simulcast subsystem, which employs one simulcast control channel 915 and one or more simulcast voice channels 925 to provide trunking communication between radio users 905 within the subsystem 630. Each site 900 in the simulcast subsystem 630 includes a plurality of repeaters 910 (one to provide active or standby control channel functionality, or operations, and additional repeaters 910 for each voice channel provided in the subsystem 630), and at least one subsystem controller 920.

[0052] To provide redundancy, a single subsystem controller 920 located at one of the sites 900 is active for the entire subsystem 630, and the remaining subsystem controllers 920 located at the remaining sites 900 serve as standby. Additionally, a single repeater 910 located at one of the sites 900 is active to provide control channel functionality, or operations, and a single repeater 910 located at each of the remaining sites 900 is provided as standby in the event of failure of the active control channel repeater 910. The remaining repeaters 910 located at each of the sites are generally designated as voice channels for each of the channels provided by the subsystem 630, however, each

repeater is also capable of performing voting and simulcast operations as explained below.

[0053] The trunked simulcast subsystem 630 employs both voting and simulcast operations to provide communication across the subsystem 630. In a simulcast operation, a single channel is usually provided by a collection of repeaters 910 distributed across multiple geographic sites comprising the subsystem (e.g., sites 900A, 900B, and 900C), wherein the repeaters 910 operate on the same frequency pair (transmit and receive), under voted and simulcast configurations, to expand the coverage area of the subsystem 630 into the sites (e.g., sites 900A, 900B, and 900C). In other words, in some embodiments, a single channel may be provided by one repeater 910 at each of the sites 900A-900C in the subsystem 630.

[0054] In a traditional simulcast LMR system, the capability of a radio's communication to reach the prime site can be limited by the transmit power of the radio. One way to improve the talkback capability of the radios is to use receiver voting to determine the location (e.g., site) of the radio to determine the best means for communicating with the radio. FIG. 10 provides an example illustration of a traditional architecture for providing receiver voting. Traditionally, receiver voting is performed by placing a number of additional radio receivers (towers) 1010 in strategic locations within the simulcast system 1015 to receive the RF signal 1020 from a transmitting radio 1025. Each tower 1010 then sends the received signal and signal strength data 1035 to a voter comparator 1030. The voter comparator 1030 compares the signal strength of each received signal and selects the best signal 1040 to use for communication. The voter comparator 1030 then sends the best signal 1040 to the subsystem controller at the prime site for further processing. By increasing the number of radio receivers within a subsystem, the overall system talkback coverage area may be expanded.

[0055] In traditional systems, such as that illustrated in FIG. 10, there is one voter comparator per channel, and the voter comparator is located at a prime site or at the central network controller in a centralized LMR architecture, which may not be the same site at which the transmitting radio is located. In these traditional implementations, voter comparators present a single point of failure for a channel.

[0056] In accordance with an embodiment of the present disclosure, the single voter comparator is eliminated (as is the central network controller), and the voter comparator functionality is integrated into each repeater in the subsystem. For example, FIG. 11 illustrates an example of such an embodiment wherein repeaters 1110 located at different sites 1115 (similar to the repeaters 910 located at the sites 900 in FIG. 9) throughout the subsystem 1130 have integrated voter comparator functionality. Although each of the repeaters 1110 in the subsystem 1130 have voter comparator functionality, the voter comparator functionality of only one of the repeaters 1110 (for a particular channel) is configured to be active at any given time. In some embodiments, the repeaters and subsystem controllers communicate with each other to determine which repeater is active. Additionally, in some embodiments, the network management system may configure a particular repeater to be active.

[0057] The repeater 1110 with active voter comparator functionality may perform voter comparator operations, including voting of signals 1120 for all sites in the subsystem 1130 (for the particular channel assigned to the repeater

1110). The voter comparator functionality of the remaining repeaters 1110 for the channel are on standby in case of failure of the active repeater 1110. This redundancy reduces the potential of operational downtime because, if the voter comparator functionality of one repeater 1110 fails, the voter comparator functionality of another repeater 1110 will become active.

[0058] If an adequate number of frequencies are not available for communication in an LMR system, a simulcast operation may be performed to reuse frequencies and cover a large geographic area. Referring again to FIG. 9, a simulcast channel may utilize several geographically separated repeaters 910 which transmit simultaneously on the same frequency, thereby reducing the number of frequencies needed for the entire subsystem 630. For example, repeaters 910A, 910B, and 910C located at sites 900A, 900B, and 900C, respectively, may all transmit simultaneously on the same frequency to provide simulcast communication between radios in the subsystem 630.

[0059] In a traditional simulcast LMR system, such as that illustrated in FIG. 12, a simulcast channel uses a simulcast controller 1210, which synchronizes the launch time (transmit time) of calls transmitted to sites 1220 in the system 1225. The simulcast controller 1210 receives the audio signal 1215, assigns a launch time, and then sends the signal 1215 to each transmitter/repeater 1230. Each transmitter 1230 transmits the signal at its respective site 1220 pursuant to the launch time, and radios 1235 communicating on that channel receive the signal from the multiple transmitters 1230. The timing system implemented by the simulcast controllers 1210 and transmitters 1230 synchronizes the launch time in the transmitter 1230 so that calls are transmitted simultaneously from all sites 1220 for a given repeater channel. This synchronization ensures that the transmission on the same frequency is in phase, thereby reducing interference.

[0060] In traditional simulcast LMR systems, such as that illustrated in FIG. 12, a single simulcast controller 1210 is provided for each channel. The simulcast controller 1210 is typically located at a prime site or at the central network controller in a centralized LMR architecture. Accordingly, the single simulcast controller 1210 in a traditional simulcast LMR system provides a single point of failure for a channel.

[0061] In accordance with an embodiment of the present disclosure, the single simulcast controller is eliminated (as is the central network controller), and the simulcast controller functionality is integrated into each repeater in the subsystem. For example, FIG. 13 illustrates an example of such an embodiment wherein repeaters 1310 located at different sites 1315 (similar to the repeaters 910 located at the sites 900 in FIG. 9) throughout the subsystem 1330 have integrated simulcast controller functionality. Although each of the repeaters 1310 in the subsystem 1330 have simulcast controller functionality, the simulcast controller functionality of only one of the repeaters 1310 is configured to be active at any given time. In some embodiments, the repeaters and subsystem controllers communicate with each other to determine which repeater is active. Additionally, in some embodiments, the network management system may configure a particular repeater to be active.

[0062] Referring again to FIG. 9, the repeater 910 with active simulcast controller functionality may perform simulcast controller operations across all sites 900 in the subsystem 630. In some embodiments, one repeater 910 may

provide simulcast controller functionality for a particular channel while other repeaters 910 allocated to that channel are on standby to provide redundancy. In other words, one repeater 910 for each channel may have active simulcast controller functionality, and the simulcast controller functionality of the remaining repeaters 910 in the subsystem 630 are on standby in case of failure of the active repeater 910 for the standby repeater's respective channel. This redundancy reduces the potential of operational downtime because, if the simulcast controller functionality of one repeater 910 fails, the simulcast controller functionality of another repeater 910 will become active.

[0063] For example, in one embodiment, repeaters 910A, 910B, and 910C are allocated to a particular channel. Repeater 910A may be active to provide simulcast controller operations for the channel allocated to repeaters 910A, 910B, and 910C, and repeaters 910B and 910C are on standby. If repeater 910A fails, repeater 910B or repeater 910C may become active to provide simulcast controller functionality for the channel allocated to repeaters 910A, 910B, and 910C.

[0064] It should be appreciated that other variations and embodiments may be considered within the scope of the present disclosure. For example, in some embodiments, one repeater 910 may provide active simulcast controller functionality for more than one channel in the subsystem 630. In this embodiment, the active simulcast repeater 910 may provide simulcast controller functionality for some, or all, of the channels in the subsystem 630. For example, repeater 910A may be allocated to a first channel, repeater 910B allocated to a second channel, and repeater 910C allocated to a third channel. Repeater 910A may provide active simulcast controller functionality for the first channel, second channel, third channel, or any combination thereof, and the remaining repeaters 910B and 910C may operate in standby mode.

[0065] In accordance with an embodiment of the present disclosure, the trunked simulcast subsystem 630 may operate in accordance with the following example call sequence discussed with reference to FIG. 9. A radio 905A initiates a call through the control channel 915. The active subsystem controller 920 (e.g., subsystem controller 920 in site 900A) processes the user-initiated call and assigns a voice channel 925 for the call. The radio users 905 communicate with each other over the voice channel 925. Any request from the initiating user 905A is received by one or more repeaters 910 of the channel, and the repeater 910 with active voter comparator functionality (e.g., repeater 910A) then performs the voter comparator operation to select the best signal. Additionally, any data packet sent to the initiating user 905A or other radio users 905 is sent to all repeaters 910 of the channel 925 through the repeater 910 with active simulcast controller functionality (e.g., repeater 910A). The repeaters 910 then simultaneously transmit the data to the radios 905 over the channel 925. In the embodiment discussed herein, both the voter comparator and simulcast controller functionality is provided by a single repeater (910A). However, it should be appreciated that, in some embodiments, the voter comparator functionality and simulcast controller functionality may be provided by separate repeaters.

[0066] As discussed above and illustrated in FIG. 14, traditional simulcast LMR systems 1400 include a prime site 1410, which hosts voter comparators, simulcast controllers, and a simulcast subsystem controller, thereby providing

simulcast synchronization and voter comparator functionality for all channels in the system 1400. The traditional simulcast LMR system 1400 embodies a centralized architecture, wherein the prime site 1410 acts as a central network controller for all sites 1420 in the system 1400. Thus, each site 1420 relies on the constant availability of the prime site 1410, and is therefore susceptible to loss of the simulcast and voter comparator functionality in the event of prime site failure.

[0067] Referring now to FIG. 15, and in accordance with the forgoing discussion of the present disclosure, the trunked simulcast subsystem 1500 of the distributed LMR architecture eliminates the need for a dedicated prime site or prime site controller. The availability of simulcast controller functionality and voter comparator functionality in each repeater enables the distribution of the prime site functionality of each channel to different sites 1510 in the subsystem 1500. Accordingly, any repeater may provide prime site functionality/operations for a channel, wherein providing prime site functionality/operations for a channel includes providing simulcast controller functionality (i.e., performing simulcast controller operations) and/or providing voter comparator functionality (i.e., providing voter comparator operations) for the channel. Prime site functionality of each repeater may be assigned through configuration, or dynamically. For example, as shown in FIG. 15, a first site 1510A may provide prime site functionality for a first channel, a second site 1510B may provide prime site functionality for a second channel, a third site 1510C may provide prime site functionality for a third channel, and a fourth site 1510D may provide prime site functionality for a fourth channel. Other repeaters in the subsystem 1500 are configured as standbys and become active upon failure of the primary repeater (i.e., the repeater providing active voter comparator/simulcast controller functionality).

[0068] As illustrated in FIGS. 9 and 15, the distributed simulcast architecture also eliminates the need for a central controller to process calls. Instead, the trunked simulcast subsystem controller (920) performs control for all calls. If the primary subsystem controller fails, one of the standby subsystem controllers becomes active, thereby providing redundancy and improved reliability in comparison to a centralized architecture.

[0069] Referring briefly to FIG. 6, the dispatch centers 650 include dispatch consoles connected to other components/subsystems in the distributed LMR architecture system 600. Each console is peer to other consoles in the system 600 and is fully distributed, thereby ensuring that the impact of a console failure is localized to the corresponding dispatch center 650, and does not affect the remaining components/subsystems in the system 600.

[0070] Reference is now made to FIGS. 16-18, which are provided in support of the following description of various redundancy advantages provided by the disclosed distributed simulcast LMR architecture. Specifically, the disclosed distributed simulcast architecture provides redundancy in at least three aspects: (i) simulcast controller and voter comparator redundancy, (ii) network failure redundancy, and (iii) site redundancy.

[0071] FIG. 16 illustrates an example embodiment demonstrating simulcast controller and voter comparator redundancy in the disclosed distributed simulcast LMR architecture. As illustrated in the example embodiment in FIG. 16, simulcast controller functionality and voter comparator

functionality is shown distributed between multiple simulcast sites 1615 in a distributed simulcast subsystem 1610. In the subsystem 1610A, sites 1615A, 1615B, 1615C, and 1615D provide prime site functionality for channels one, two, three, and four, respectively, and site 1615D also provides the active subsystem controller. In subsystem 1610B, the repeater providing the simulcast controller/voter comparator functionality at site 1615A fails, and the prime site functionality for channel one changes to one of the standby repeaters located at one of the other sites. In this example, site 1615C assumes prime site functionality for channel one. As shown in the subsystem 1610B, site 1615C provides prime site functionality for channels one and three. In the embodiment illustrated in FIG. 16, the subsystem controller location at site 1615D remains unchanged.

[0072] FIG. 17 illustrates an example embodiment demonstrating network failure redundancy in the disclosed distributed simulcast LMR architecture. In FIG. 17, a trunked simulcast subsystem 1700 includes sites 1715A, 1715B, 1715C, and 1715D providing prime site functionality for channels one, two, three, and four, respectively, and site 1715D also providing the active subsystem controller for the subsystem 1700. As illustrated in FIG. 17, if a network failure breaks the trunked simulcast subsystem 1700 into halves 1710 and 1720, each half becomes a smaller simulcast subsystem.

[0073] One of the subsystem controllers in each of the smaller subsystems 1710 and 1720 becomes active and continues to provide user communication on a reduced number of channels. For example, in the first reduced subsystem 1710, the subsystem controller at site 1715A becomes active and provides call control functionality for sites 1715A and 1715C. In the second reduced subsystem 1720, the subsystem controller at site 1715D remains active and provides call control functionality for sites 1715B and 1715D. This built-in redundancy ensures that users in each half 1710 and 1720 can still communicate with each other without interfering with RF signals in the overlapping area.

[0074] FIG. 18 illustrates an example embodiment demonstrating site redundancy in the disclosed distributed simulcast LMR architecture. In FIG. 18, a trunked simulcast subsystem 1800 includes sites 1815A, 1815B, 1815C, and 1815D providing prime site functionality for channels one, two, three, and four, respectively, and site 1815A also providing the active subsystem controller for the subsystem 1800. Site redundancy is enabled by the combination of redundant simulcast controller functionality and voter comparator functionality in each repeater and a redundant subsystem controller at each site 1815. Therefore, in case of a catastrophic failure of a site 1815, the rest of the sites 1815 continue to provide coverage with little impact on the subsystem 1800. For example, as shown in FIG. 18, if site 1815A experiences a catastrophic failure, one of the standby repeaters located at one of the other sites (in this example, site 1815C) is activated and assumes prime site functionality for channel one. Additionally, one of the standby subsystem controllers located at one of the other sites (in this example, site 1815D) is activated and becomes the subsystem controller for the subsystem 1800.

[0075] When compared to centralized LMR system architecture and traditional simulcast LMR systems, the foregoing disclosure of the distributed simulcast architecture provides various advantages and benefits. For example, the disclosed system provides increased reliability because the

removal of a prime site eliminates the single-point-of-failure structure provided in a traditional simulcast system. Furthermore, distributing the functionality of the prime site to the various sites and equipment comprising the distributed simulcast subsystem reduces costs and maintenance required to maintain the system. Additionally, providing a subsystem controller at each site in the subsystem offers multiple levels of redundancy of the controller, and affords communication throughout the subsystem even in the event of various failures. Finally, providing voter comparator and simulcast controller functionality in each repeater provides N times the voter/simulcast controller availability in a traditional simulcast system, where N represents the number of sites in the subsystem. This also allows redundancy of voter comparator functionality and simulcast controller functionality within a site or across multiple sites (to survive network failure, site failure, or equipment failure), thereby providing communication in the event of multiple failures, and providing automatic and dynamic tuning of transmission launch time.

[0076] A number of additional and alternative embodiments of the disclosed system and method may be provided without departing from the spirit or scope of the present disclosure as set forth in the claims provided herein. These various embodiments are believed to be understood by one of ordinary skill in the art in view of the present disclosure.

What is claimed is:

1. A system for providing communication in a distributed land mobile radio (LMR) system architecture, the system comprising:

a plurality of comparators, at least one of the plurality of comparators configured to receive at least one signal from a device,

wherein at least one of the plurality of comparators is an active comparator configured to receive a plurality of signals from the plurality of comparators and transmit a signal of the plurality of signals that has a greatest signal strength, and

wherein at least one of the plurality of comparators is a standby comparator configured to switch to an active mode from a standby mode, the standby comparator being distinct from the active comparator.

2. The system of claim 1, wherein the plurality of signals are radio frequency signals.

3. The system of claim 1, wherein the active comparator is an active voter comparator configured to perform voter comparator operations.

4. The system of claim 1, wherein the standby comparator is configured to switch to the active mode from the standby mode upon failure of the active comparator.

5. The system of claim 1, wherein at least two of the plurality of comparators communicate to determine which of the plurality of comparators is the active comparator.

6. The system of claim 1, further comprising a network management system configured to select which one of the plurality of comparators is the active comparator.

7. A system for providing communication in a distributed land mobile radio (LMR) system architecture that includes a plurality of devices configured to communicate with one another via the system, the system comprising:

a plurality of controllers, at least one of the plurality of controllers configured to control communication between the plurality of devices,

wherein at least one of the plurality of controllers is an active controller configured to provide communication in the LMR system architecture,

wherein providing communication in the LMR system architecture comprises controlling transmission of a plurality of signals between the plurality of devices, and

wherein at least one of the plurality of controllers is a standby controller configured to switch to an active mode from a standby mode, the standby controller being distinct from the active controller.

8. The system of claim 7, wherein at least one of the plurality of devices comprises one of an LMR, a mobile radio, a subscriber unit, a repeater, a radio transceiver, a radio tower, a fire truck, an ambulance, a police vehicle, or an emergency vehicle.

9. The system of claim 7, wherein at least one of the plurality of controllers is configured to generate a communication channel.

10. The system of claim 9, wherein at least one of the plurality of controllers is configured to transfer one or more calls from at least one device of the plurality of devices to a third controller via the communication channel.

11. The system of claim 10, wherein the standby controller is configured to switch to the active mode from the standby mode upon failure of the active controller.

12. The system of claim 10, wherein at least two of the plurality of controllers operate at a same frequency.

13. A system for providing communication in a distributed land mobile radio (LMR) system architecture that includes a plurality of devices configured to communicate with one another via the system, the system comprising:

a plurality of repeaters, at least one of the plurality of repeaters configured to selectively receive a signal of a plurality of signals from one or more of the plurality of devices and assign a launch time to the signal of the plurality of signals; and

a transmitter configured to transmit at least one signal of the plurality of signals to a controller that is configured to control transmission of the at least one signal of the plurality of signals from one or more of the plurality of devices based on the launch time of the signal of the plurality of signals.

14. The system of claim 13, wherein at least one repeater of the plurality of repeaters is an active repeater configured to initiate a simulcast controller operation.

15. The system of claim 14, wherein the simulcast controller operation comprises synchronizing a transmit time of calls transmitted by the active repeater.

16. The system of claim 15, wherein the simulcast controller operation further comprises transmitting a radio call at least substantially simultaneously.

17. The system of claim 13, wherein the controller is configured to provide a communication channel.

18. The system of claim 17, wherein at least one repeater of the plurality of repeaters is an active repeater configurable to perform a simulcast controller operation for the communication channel.

19. The system of claim 18, wherein the active repeater is configurable to perform the simulcast controller operation when the active repeater operates in an active mode.

20. The system of claim **19**, wherein, while the active repeater is in the active mode, at least a portion of the plurality of repeaters operate in a standby mode.

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